INCH-POUND

MIL-M-28787/111A(NW)
AMENDMENT 3
16 December 1992
SUPERSEDING
AMENDMENT 2
14 February 1986

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

TERMINATOR, INTERFACE, DIGITAL, KEY CODE MDN

This amendment forms a part of Military Specification MIL-M-28787/111A(EC), dated 10 December 1981, and is approved for use by the Crane Division, Naval Surface Warfare Center, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

TABLE I, Test conditions A1-A10, Test type column: Delete "X" and substitute "S".

TABLE I, Test condition B1, Max column: Delete "2" (4 places) and substitute "1" (4 places).

PAGE 5

*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through F shall be performed in any order with the following exceptions. Test conditions A, B2, and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, or F are part of a sequence of tests, the order of testing shall be as follows: A, B, and F where applicable, followed by the remaining tests in the sequence."

AMSC N/A FSC 5963
DISTRIBUTION STATEMENT A Approved for public release distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:04Z
Check the source to verify that this is the current version before use.

4.5.1: Delete and substitute the following:

"4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through F in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B1, C, D, and E shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2."

4.5.3: Delete and substitute the following:

"4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B1, C, D, and E shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B1, C, D, and E shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I."

6.3: Delete and substitute the following:

"6.3 Cross reference of classification. Military Specification MIL-M-28787/111 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658834. Part number M28787/111-1 supersedes part numbers 2658834 and M28787/111."

PAGE 9

TABLE III, Test condition B1, Max column: Delete "2" (1 place) and substitute "1" (1 place).

PAGE 11

*TABLE V: Delete and substitute new table V as printed on page 4 of this amendment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:04Z
Check the source to verify that this is the current version before use.

*FIGURE 3: Delete and substitute new figure 3 as printed on page 5 of this amendment.

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy - NW

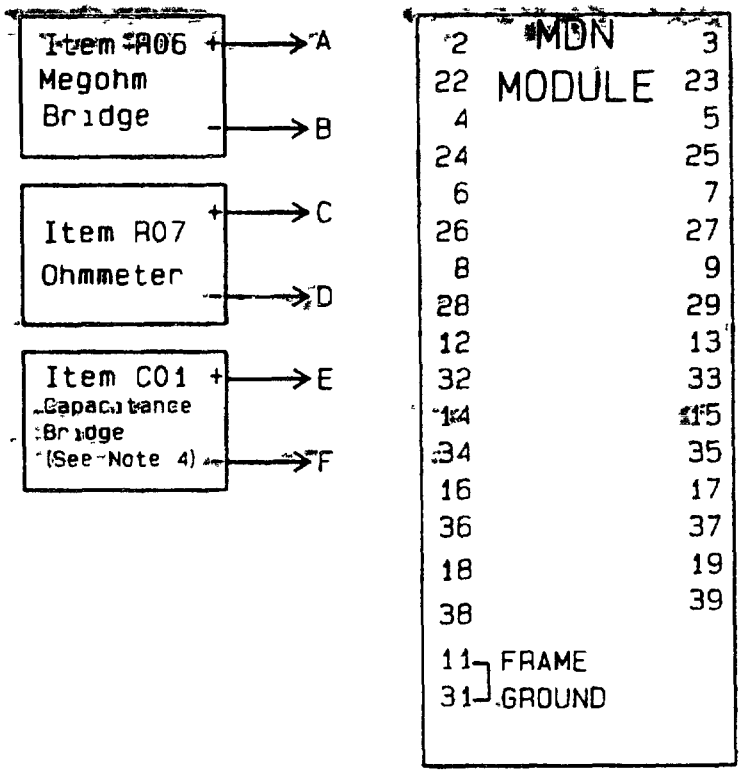
(Project 5963-N855)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:04Z
Check the source to verify that this is the current version before use.

TABLE V. Capacitance test

Test cond	Parameter	Connect		Initial limits 25 ± 5°C ambient		Unit	Test fig.
		E to pin	F to all pins except	Min	Max		
D1	Capacitance	2	3	0	30	pF	3
D2		22	23				
D3		4	5				
D4		24	25				
D5		6	7				
D6		26	27				
D7		8	9				
D8		28	29				
D9		12	13				
D10		32	33				
D11		14	15				
D12		34	35				
D13		16	17				
D14		36	37				
D15		18	19				
D16	Capacitance	38	39	0	30	pF	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:04Z
Check the source to verify that this is the current version before use.



WARNING:
When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTES:
1. For test condition A1, the megohm bridge voltage shall be 100 ± 10Vdc.
2. For test conditions A2-A10, the megohm bridge voltage shall be 30 ± 3Vdc.
3. For test condition B2, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of item R07.
4. For test conditions D1-D16:
(a) Test fixture and lead capacitance must be measured for each test condition prior to insertion of module into fixture and then subtracted from the capacitance measured with the module inserted. Test fixture and lead capacitance must not exceed 30 pF.

(b) Equivalent series capacitance shall be measured at a frequency of 1 kHz and a bridge voltage of 0.3 Vrms maximum.

FIGURE 3. Module integrity, continuity, resistance, and capacitance test circuit for tables II, III, IV, and V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:04Z
Check the source to verify that this is the current version before use.